

Auteur: Luca Letroye
Studierichting: Informatica
Oefeningenreeks 5G nr. 9.

$$y^2 = x^4 + x^5 = x^4(1+x)$$

$$x^4(1+x) \geq 0 \text{ als } x \geq -1$$

$$x^4(1+x) < 0 \text{ als } x < -1$$

$$\begin{aligned} V &= \pi \int_{-1}^0 (x^4 + x^5) dx \\ &= \pi \left(\int_{-1}^0 x^4 dx + \int_{-1}^0 x^5 dx \right) \\ &= \pi \left(\left[\frac{x^5}{5} \right]_{-1}^0 + \left[\frac{x^6}{6} \right]_{-1}^0 \right) \\ V &= \pi \left(\frac{1}{5} - \frac{1}{6} \right) = \frac{\pi}{30} \end{aligned}$$

5G-9-luca.letroye.INF.png

References